5b. talk2me

Let's set up a **serial communication** between our Arduino and the computer to monitor changing values.

In the code below, we'll map the range 0–1023 to a custom range of 10–500, send both variables over the serial port, and use the Arduino Serial Monitor to view them. Click the Serial Monitor button in the toolbar and select the same baud rate specified in the `Serial.begin()` call.

The circuit remains unchanged.

Code

```
int sensorPin = A0;    // select the input pin for the potentiometer
int ledPin = 2;        // select the pin for the LED
int sensorValue = 0;   // variable to store the value coming from the sensor
int outputValue = 0;   // variable to store a scaled value of the sensorvalue

void setup() {
  // declare the ledPin as an OUTPUT
  pinMode(ledPin, OUTPUT);
  // initialize serial communications at 9600 bps
  Serial.begin(9600);
}

void loop() {
  // read the value from the sensor:
  sensorValue = analogRead(sensorPin);

  // map or scale it to a custom range:
  outputValue = map(sensorValue, 0, 1023, 10, 500);

  // print the results to the Serial Monitor:
  Serial.print("sensor = ");
  Serial.print(sensorValue);
  Serial.print("\t output = ");
  Serial.println(outputValue);

  // turn the ledPin on
  digitalWrite(ledPin, HIGH);
  // stop the program for <sensorValue> milliseconds:
  delay(sensorValue);
  // turn the ledPin off:
  digitalWrite(ledPin, LOW);
  // stop the program for for <sensorValue> milliseconds:
  delay(sensorValue);
}
```